

Question

Gas **A**, which is very important to industry and agriculture, has rich chemical properties. Before World War I, German chemist Haber optimized the conditions for the direct reaction of elements to produce **A**, making it possible for large-scale industrial production. Electrolysis of HF salts of **A** and HF melts can yield a colorless and odorless stable gas **B**. In addition to this method, first used by the German OttoRuff group, controlled fluorination of **A** over a copper catalyst can also produce **B**; however, if the temperature is not properly controlled, the generated **B** will fluorinate the copper catalyst and transform itself into **C**. At room temperature and above, **C** easily dissociates and can be regarded as a dimer of a certain free radical. In 1966, two research groups independently discovered gas **D**. One method of preparation is the discharge oxidation of **B** with oxygen at $-196\text{ }^{\circ}\text{C}$; the other is the reaction of **E** with the single-electron oxidant IrF_6 , where the byproduct **F** contains diatomic cations and octahedral coordinated anions. The cation of **F** can be reduced to gas **G**, which is easily oxidized in air to **H** and reaches equilibrium with the dimer of **H**, exhibiting a reddish-brown color. Direct fluorination of **G** or reaction of **G** with AgF_2 can yield the substrate **E** used to generate **D**. Direct fluorination of **B** in the presence of SbF_3 can yield salt **I**, whose cation is isoelectronic with **D**.

Deduce the substances represented by the above symbols and judge whether the following statements are correct or not.

1. The molecular weight of **C** is greater than 100
2. The molecular weight of **A** is greater than 20
3. **D** and **E** contain the same elements
4. **F** contains three elements
5. **G** is paramagnetic
6. The dipole moment of **A** is less than **B**
7. The cation of **I** has T_d symmetry

M is the sum of the squares of the serial numbers corresponding to all correct statements, and N is the square of the sum of the serial numbers corresponding to all incorrect statements. Calculate the value of M/N and select the option with an error within 0.001 in the options.

A. All other options are incorrect

B. 0.491

C. 0.420

D. 0.583

E. 0.085

F. 3.333

G. 0.785

Answer

Correct Answer: **D**

Detailed Explanation

The method of producing **A** from elements was optimized by German chemist Haber before World War I, and **A** is used in industry and agriculture, so **A** is NH_3 . The molecular weight of ammonia is 17, so statement 2 is incorrect.

CHECKPOINT

1 PTS

A is NH_3

B is obtained by electrolysis of the HF salt of ammonia, i.e., ammonium fluoride, or by fluorination of ammonia, indicating that it is a fluorine-substituted ammonia. **B** will convert itself into **C** with copper fluoride catalyst, and since **C** is easily dissociated and can be regarded as a dimer of free radicals, **C** is N_2F_4 . The dissociated NF_2 free radical is stabilized by the strong electron-withdrawing effect of fluorine and the conjugation of lone pair electrons. **C** is the product of **B** fluorinating the catalyst, so **B** is NF_3 .

CHECKPOINT

1 PTS

B is NF_3

CHECKPOINT

1 PTS

C is N_2F_4

The molecular weight of **C** is 104, which is greater than 100, so statement 1 is correct.

The lone pair electrons of NH_3 are in the same direction as the bond moment, while NF_3 is the opposite, so the dipole moment of **A** is greater than that of **B**, so statement 6 is incorrect.

CHECKPOINT

1 PTS

Statement 6 is incorrect

D is obtained by oxidizing **B** with oxygen, which can only be ONF_3 .

CHECKPOINT

1 PTS

D is ONF_3

The cation of **F** can be reduced to the gas **G**, and **G** is easily oxidized in the air to **H**, and is in equilibrium with the dimer, showing reddish-brown color, so **H** is NO_2 , and **G** is NO .

CHECKPOINT

1 PTS

G is NO

CHECKPOINT

1 PTS

H is NO_2

NO has a single electron and is a paramagnetic gas, so statement 5 is correct.

Then the cation of **F** is the nitrosyl cation, which conforms to the description of a diatomic cation. Since IrF_6 is a single-electron oxidant, and its oxidation of **E** can generate ONF_3 , and at the same time generate **F** as a byproduct, the anion of **F** is IrF_6^- , which conforms to the description of the octahedral coordinated anion in the question, so **F** is $[NO][IrF_6]$.

CHECKPOINT

2 PTS

F is $[NO][IrF_6]$

F contains 4 kinds of elements, so statement 4 is incorrect.

Fluorinating NO can obtain **E**, indicating that **E** is a fluoride or oxyfluoride of N . And **E** can be oxidized to ONF_3 by IrF_6 , indicating that it is the oxyfluoride ONF .

CHECKPOINT

1 PTS

E is ONF

D and **E** contain the same elements, so statement 3 is correct.

Direct fluorination of NF_3 in the direct presence of SbF_3 yields the salt **I**. From common sense, the anion is SbF_6^- , and the cation contains only nitrogen and fluorine, so it can only be NF_4^+ . Therefore, **I** is $[NF_4][SbF_6]$. This conforms to the description that its cation and ONF_3 are isoelectronic.

CHECKPOINT

1 PTS

I is $[NF_4][SbF_6]$

Therefore, the cation of **I** has T_d symmetry, so statement 7 is correct.

CHECKPOINT

1 PTS

Statement 7 is correct

In summary, the correct statements are 1, 3, 5, 7, and the incorrect statements are 2, 4, 6.

$$M = 1 + 9 + 25 + 49 = 84$$

$$N = (2 + 4 + 6)^2 = 144$$

$$M/N = 0.583$$

Therefore, choose D